

Reproducible analyses in R

What, Why, and How?

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Topics

- Open Science Practices
- Reproducibility: What it is and why/how to practice it
- Concepts for building a reproducible workflow

What is Open Science?

“Open science” is an umbrella term used to refer to the concepts of openness, transparency, rigor, reproducibility, replicability, and accumulation of knowledge, which are considered fundamental features of science”

— Crüwell et al. (2019), p.3

- a movement developed to respond to crisis in scientific research
 - lack of accessibility, transparency, reproducibility, and replicability of previous research
- transparency is key to all facets of Open Science
 - it allows for full evaluation of all stages of science
- Open Access, software, data, code, materials...

Systemic problem in science

- the combination of
 - publication bias
 - * journals favour novel, significant findings
 - publish or perish
 - * researchers' careers depend on publications

- can/does/did lead to:
 - HARKing
 - * Hypothesising After Results are Known
 - p-hacking
 - * (re-)running analyses until a significant effect is found
 - replication crisis
 - * pervasive failure to replicate previous research

How to practice Open Science

- Figure 1 shows some suggestions from Kathawalla et al. (2021)
- Open Science is not all-or-nothing
- there are things I consider the bare minimum
 - detailed experiment plan, ideally public
 - openly available materials (e.g., stimuli)
 - share code and data
- the important thing is to do what you can

💡 Which Open Science research practices in Figure 1 do you already practice? Are there any you'd like to start implementing?

i The replication crisis

The *replication crisis* refers to the failure of many replication studies to replicate the findings of influential studies. The result of this “crisis” is a move towards Open Science Practices, which emphasise transparency along all stages of research (conception, planning, data collection, data cleaning, data analysis, reporting).

The issue became more widespread with the publication of Ioannidis (2005), entitled *Most published research findings are false*. This paper defined bias in terms of design, analysis, and presentation factors with a focus on issues with p -values and statistical power. Since then there have been many replication attempts of influential (and lesser influential) papers. Strikingly, Open Science Collaboration (2015) reports 100 psychological studies run by 270 collaborators. They reported significant effects in only 36% of replications, with 47% of originally reported effects falling within 95% CIs of the replication effect. In essence: fewer significant findings and smaller effect sizes were found in replication studies compared to the original 100 studies Figure 2.

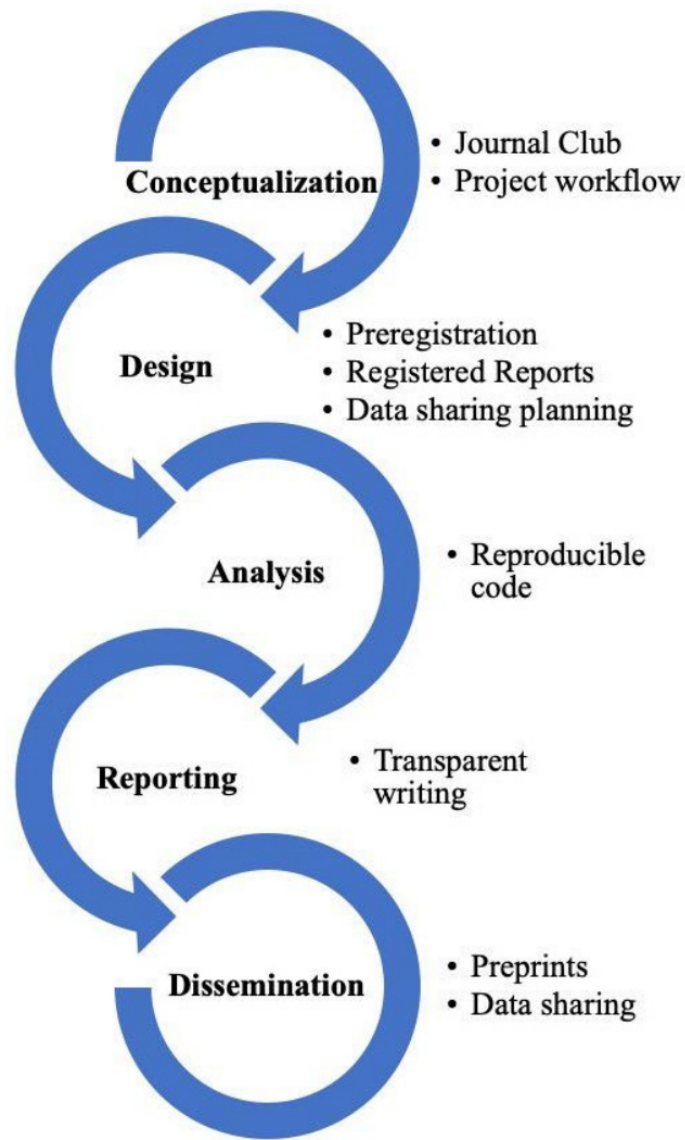


Figure 1. Open Science research practices across the research cycle

Figure 1: Image source: Kathawalla et al. (2021) (all rights reserved)

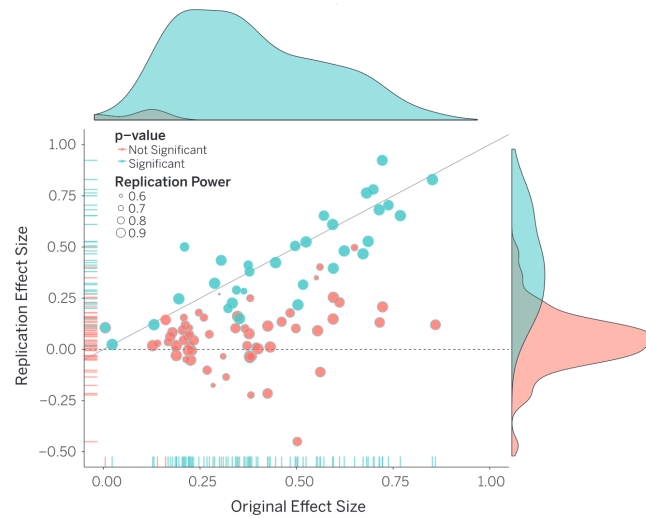


Figure 2: Source: Open Science Collaboration (2015) (all rights reserved)

Reproducibility

What is reproducibility?

- one piece of the Open Science pie
- generating the same results with the same data and analysis scripts
- seems obvious, but requires organisation and forethought before and during data collection/analysis
- bare minimum: share the code and the data (Laurinavichyute et al., 2022)

Reproducibility vs. replication

- the two terms have been used interchangeably in the past (e.g., in the title of Open Science Collaboration, 2015)
 - we'll define them as follows (and this is becoming the standard distinction, imo)

Reproducibility

- re-analysing the same data using (ideally) the same scripts, software...
- aim: produce the same results (means, model estimates, etc.)

- why: tests for errors, coding mistakes, biases, etc.

Replication

- re-running a previous experiment, ideally with the same materials, set-up...
 - ideally the same analysis workflow as the original study (i.e., like reproducing the analyses but with new data)
- aim: test whether results are replicated with new data in terms of direction and magnitude
- in short:
 - reproducibility = re-analysis of the *same data*
 - replication = collection of *new data*

Why implement reproducibility in my workflow?

- firstly: the help future you (or collaborators/other researchers)!
 - you may return to your analyses tomorrow, next month, or next year
- to ensure robustness and to document your steps
 - ‘researcher degrees of freedom’ and the ‘garden of forking paths’: there’s more than one way to analyse a certain dataset
 - we can try to plan ahead in detail (e.g., pre-register your analysis plan), but there will always be decisions made that were not foreseen
- lastly: it makes your life *much* easier and streamlines your workflow

How to implement reproducibility?

- not exactly straightforward
 - there are degrees of reproducibility
 - the rest of our time will be spent on this topic
- sharing code and data is a first step
 - think of the FAIR principles of data sharing
 - apply them to sharing analyses as well

Practice FAIR principles

- guidelines for sharing digital resources
- refers broadly to (meta)data, let's extend them to analysis code



Figure 3: Source: [National Library of Medicine](#) (all rights reserved)

- **findable** and **accessible**: where materials are stored
 - in *findable* repositories
 - that are *accessible*, i.e., do not require an account
- **interoperable** and **reusable**: format of data (and code)
 - the importance of future use
 - and use beyond your precise computational environment

Conduct a code review

- a great way to test the FAIR principles
 - code review!
 - i.e., have a colleague try to access your data/run your code
 - * either via an online repository
 - * or send them your project folder

The reproducibility spectrum

- reproducibility is on a continuum, referred to as the *reproducibility spectrum* in Peng (2011) (Figure 4)
 - *linked* means “all data, metadata, and code [is] stored and linked with each other and with corresponding publications” (Peng, 2011, p. 1227)
 - *executable* is not explained, and is more difficult to guarantee long-term as it depends on software versions
 - but at minimum we can assume it refers to code running on someone else’s machine

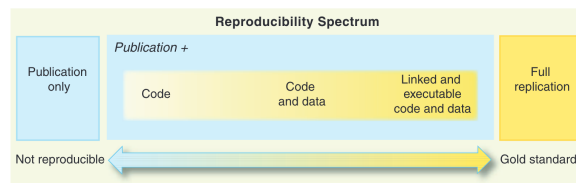


Fig. 1. The spectrum of reproducibility.

Figure 4: Source: Peng (2011)

Share the code, not just the data

- Why?
 - key details are often missing from ‘Methods’ sections
- suggestions for researchers from Laurinavichyute et al. (2022)
 1. Share data in usable form
 - with pre-processing code
 2. Use publicly accessible repositories
 - e.g., OSF

3. Use non-proprietary data formats
 - e.g., not `.xls` files (Excel)
4. Provide documentation
 - e.g., README, data dictionaries
5. Share code *and* data
 - they estimate a 38% increase in reproducibility
6. Teach data management and computing skills
 - that's what this workshop is for!

Data and code $\neq$ Reproducibility

- access to data and code does not mean analyses are reproducible
 - what can go wrong? Examples from Laurinavichyute et al. (2022)
1. Data problems
 - inaccessible data
 - incomplete data (e.g., 2/3 experiments)
 2. Code problems
 - incomplete code
 - error messages
 - code rot: outdated syntax or environment
 - proprietary software
 3. Documentation problems
 - data difficult to interpret
 - no README file/data dictionary
 - unclear folder/file/variable naming convention
 - manuscript contradicts code
 4. Unclear terms of use
 - no licence specification

What should (ideally) be shared?

- materials
 - protocols
 - stimuli
 - experiment set-up
- documentation
 - README
 - metadata
- data
 - raw
 - * e.g., text files, audio, video, or images
 - processed
- analysis code
 - pre-processing
 - analyses
- materials are helpful for replication
 - but also for inspection of e.g., design
- data and code are necessary for reproducibility
 - along with proper documentation of software used

Reproducibility rates of published works

- rates of reproducibility vary across fields (see Bochynska et al., 2023 for a review)
 - open access: 25-65%
 - data and analyses sharing: 11-33%
 - pre-registrations: 0-3%

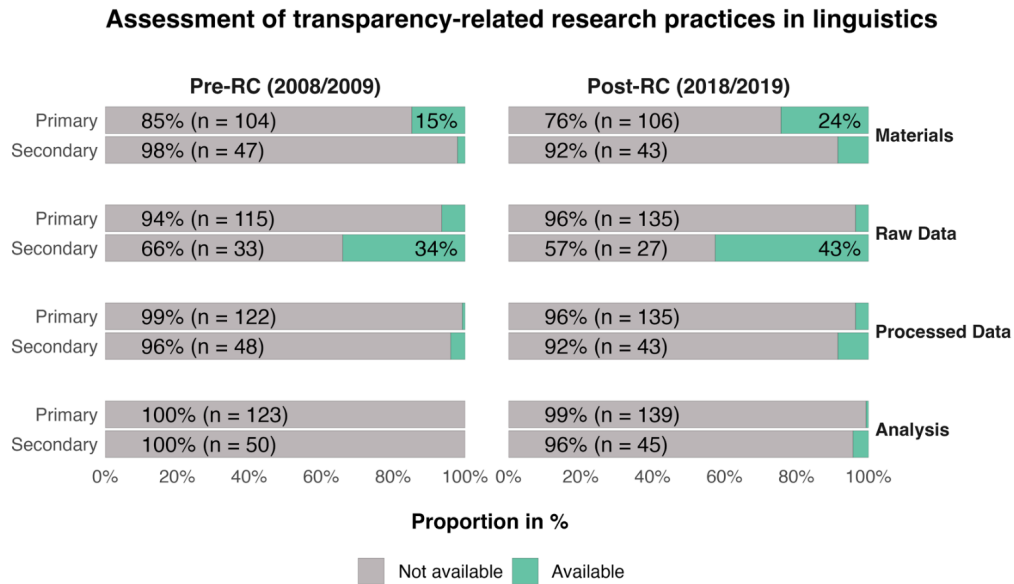


Figure 2: Percentages of the available and not available materials, raw data, processed data, and analysis scripts for the pre-RC (left) and post-RC (right) time windows, displayed separately for primary data (Primary) and secondary data (Secondary), for the empirical study articles in the sample. The Other category was excluded.

Figure 5: Source: Bochynska et al. (2023), p. 11 (all rights reserved)

Reproducibility rates in linguistic research

- meta-analysis of 519 randomly sampled articles from various linguistic journals
 - pre- and post-reproducibility crisis (2008/9, 2018/19) (Bochynska et al., 2023)
 - differentiated between primary (collected for study) and secondary (pre-existing) data
- reported a post-RC increase in shared materials, data, and analyses
 - but still low rates of each
- higher rates of secondary data sharing, presumably due to publicly available corpora
- data shared more often than analyses, pre- and post-RC

Journal of Memory and Language

- meta-analysis of articles from JML (Laurinavichyute et al., 2022)
 - before and after an Open Science Policy was introduced in 2019

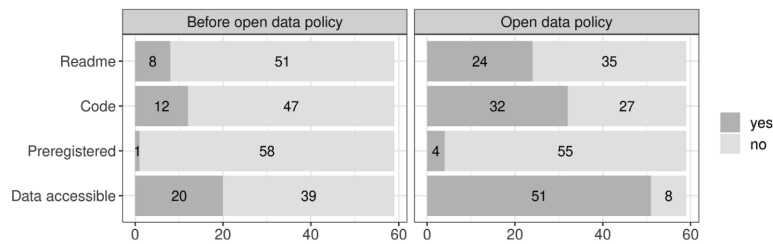


Fig. 1. A summary showing the number of papers which had a readme file or the like that provided some documentation, for which the code was available, which were preregistered, and for which the data were accessible.

Figure 6: Source: Laurinavichyute et al. (2022), p. 5 (all rights reserved)

- code and data availability improved
- but reproducibility rate ranged from 34-56%, depending on criteria
- higher rates compared to field-wide meta-analysis (Bochynska et al., 2023)

Reproducible Practices

Beyond the reproducibility spectrum

- there are different levels of reproducibility
 - the *bare minimum* is sharing the code and data

- *and* including session information:
 - * which operating system was used
 - * which software/package versions were used
- going bigger:
 - project-oriented workflow
 - project-specific filepaths
 - contained in a single project folder
- we will be using RProjects to achieve this

Project management

- folder structure
- project-relative file paths
- appropriate documentation
 - e.g., README files
- it's great to map out your project structure early on
 - but it will grow as you go along
 - reproducible principles facilitate adapting as it grows

Naming conventions

- there are some “rules” for naming files, folders, and variables
 - [The Turing Way: Naming files, folders, and other things](#)
 - [Jenny Bryan: naming things \(Reproducible Science Workshop 2015\)](#)
 - [Danielle Navarro: Project structure](#)
- 1. Avoid special characters
 - ensures machine readability
- 2. Make names concise but meaningful
 - ensures human-readability
- 3. Avoid spaces
 - try **CamelCase**, snake case (**snake_case**), or skewer case (**skewer-case**)
 - or use hyphens (-) to separate chunks, and underscores (_) to connect words of the same chunk

4. Consider default ordering

- e.g., with dates (ISO 8601): YYYY-MM-DD
- with folders or files: numerical prefixes (e.g., 01-data_cleaning.R, 02-data_visualisation.R)

5. Be *consistent*

- as long as your names are machine and human readable

💡 Tidy data

Tidy datasets are all alike but every messy dataset is messy in its own way
— Wickham (2014)

- Tidy data

Literate programming

Instead of imagining that our main task is to instruct a *computer* what to do, let us concentrate rather on explaining to *human beings* what we want a computer to do.

— Knuth (1984), p. 97

- originally used to refer to writing programs
- but also applies to analysis code
 - especially if we're aiming for reproducibility
- main concepts:
 - code is linear (this pre-dates Knuth, 1984)
 - informative but concise commenting
- main benefits:
 - facilitates maintenance
 - helpful for future-you, collaborators, etc.

Version control (not covered in this workshop)

- git: local tracking
- useful for the analysis and writing phases
 - but can be tricky for collaboration
- GitHub/GitLab: remote tracking
 - store your changes to your local git repository
 - then push them to your remote repository
- safe guards against local hardware/software issues
 - lost or damaged computer or local files
- and allows for collaboration or sharing

Writing (not covered in this workshop)

- dynamic reports with Markdown syntax
 - e.g., Rmarkdown, Quarto
 - integration of data, code, and prose
 - * facilitates cross-referencing within document
 - * integration of citation management tools
 - * supports LaTeX syntax for example sentences and tables
- **papaja** package for APA-formatted Rmarkdown documents
- challenge: collaboration
 - not all collaborators know these tools
 - track changes not currently possible

Data management and sharing

Data Management (and Sharing) Plans (DM(S)P)

- research data management is relevant for all stages of the data life cycle
 - planning, collection, processing, archiving, publishing
- DMSPs are required by some funding bodies
 - even if not, they're an important part of project planning

- questions to consider:
 - do I have data from human participants?
 - do I have data from vulnerable groups (children, patients, etc.)
 - have I collected any identifiable data from humans? (direct or indirect)

Facilitating data management/sharing

- planning and implementing folder structure, file and variable names
- keep everything relevant to a certain project in one place (i.e., folder)
 - use subfolders appropriately
 - avoid mixing subfolders and files within a single folder

Documentation

- metadata
 - project README
 - codebook/data dictionary
- README should contain
 - a project description
 - relevant links
 - description of folder structure
- can be updated as the project develops
- README.md files in GitHub/Lab are automatically used as a project description
 - .md is a plain text document
 - uses markdown syntax
- README files don't need to be markdown files, but

Version control (again)

- version control is an important aspect of data management
 - can be done with git, or manually
- manual version control ()

Persistent (public) storage

- GitHub/Lab are sub-optimal
 - developer-focused
 - typically lack thorough documentation/metadata
 - not very user-friendly for non-users
- OSF, Zenodo
 - Open Science-focused
 - can be linked to a GitHub/Lab repository
 - facilitate thorough documentation
 - user-friendly

ZAS Data Protocol

From the Employee Handbook (p. 37-8):

All data must always be stored on your personal OneDrive or a SharePoint drive of the relevant project.

Make sure this is the case!! This however does not make the data public, which usually happens after publication of a relevant manuscript.

Steps we'll take

1. Open source software:
 - R, an open source statistical programming language
 - in RStudio, an IDE (integrated developer environment)
 - with [R Projects](#)
2. Project-oriented workflow:
 - establish folder structure
 - and file/variable naming conventions
 - use project-relative filepaths with the **here** package
 - establish and maintain project-relative package library with **renv** (time permitting)
3. Practice literate programming:
 - writing clean, commented, linear code
 - in dynamic reports (e.g., Quarto, R markdown)
 - practice modularity, i.e., 1 script = 1 purpose

4. Sharing and checking our code

- uploading our code and data to an OSF repository
- conducting a code review

Topics

- Open Science Practices
- Reproducibility: What it is and why/how to practice it
- Concepts for building a reproducible workflow

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